JanGyan — Learn Without Limits

Category 6: Environment & Civic Responsibility

Module 4 - Air Quality & Pollution Control

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Air Quality & Pollution Control** — what causes air pollution, how to read and respond to the Air Quality Index, and the practical steps every household and community can take to protect themselves and reduce their contribution to the problem. Whether you are a student, a parent, a commuter, or a civic volunteer, this handbook is written so you can read it once and walk away confident and informed.

By the end of this module, you will not need to search anywhere else to understand the fundamentals of air quality management in India.

Learning Objectives

After completing this module, you will be able to:

- Define air pollution and identify its major pollutants and sources
- Read and interpret the Air Quality Index (AQI) and respond appropriately
- Distinguish between indoor and outdoor air pollution risks
- Understand the health effects of prolonged air pollution exposure
- Apply practical pollution-prevention habits in daily life
- Choose greener transportation options where feasible
- Recognise the government bodies and laws governing air quality in India
- Know exactly how and where to report a pollution violation

Why This Topic Matters

Air pollution is one of the most significant public health risks facing India today. Unlike water or soil contamination, polluted air cannot be avoided by staying indoors alone — it affects nearly everyone in a city or region, from newborns to the elderly, every single day they breathe.

> ***Reality Check:*** *In 2025, 190 out of 229 Indian cities monitored exceeded the national annual PM10 standard, and Byrnihat, Delhi, and Ghaziabad recorded the highest annual PM2.5 levels in the country (Centre for Research on Energy and Clean Air, 2026). India now operates over 1,600 air quality monitoring stations across 584 cities — but monitoring alone does not clean the air; it takes sustained action from government, industry, and citizens together.*

Much of the pollution reaching our lungs comes from sources we interact with directly — vehicle exhaust, waste burning, and construction dust. This means air quality is not only a policy issue; it is also shaped by millions of individual daily choices.

This is why JanGyan continues its Environment & Civic Responsibility category with this essential, practical module.

Introduction

Air pollution is the contamination of the atmosphere by harmful gases, particulate matter, and chemicals released from natural and human-made sources, degrading air quality to a level that harms human health, ecosystems, and the climate.

In simple words: air pollution means the air around us contains substances it should not — tiny particles and gases that, breathed in over time, damage the lungs, heart, and other organs, even when they are invisible to the naked eye.

Pollution control refers to the technologies, regulations, and behaviours used to reduce the release of these harmful substances at the source — from a factory's emission filters to a citizen's choice to avoid burning waste at home.

Fundamental Concepts

Before going deeper, let us understand a few foundational ideas that the entire subject of air quality is built upon.

Types of Air Pollutants

Air pollutants are broadly grouped by their source and physical form.

Type	Examples
Particulate Matter (PM2.5, PM10)	Dust, smoke, soot particles
Gaseous Pollutants	Carbon monoxide, sulfur dioxide, nitrogen oxides
Volatile Organic Compounds (VOCs)	Paints, solvents, fuel vapours
Ground-Level Ozone	Formed by the reaction of sunlight with other pollutants

Key Definitions

- **PM2.5** — Fine particulate matter with a diameter of 2.5 micrometres or smaller, able to penetrate deep into the lungs and bloodstream.
- **PM10** — Coarser particulate matter up to 10 micrometres in diameter.
- **NAAQS** — National Ambient Air Quality Standards, the permissible pollutant limits set by the CPCB.
- **Non-Attainment City** — A city that has consistently failed to meet NAAQS over five years, requiring a dedicated clean air action plan.
- **Source Apportionment** — A scientific study identifying which sources (vehicles, industry, dust, burning) contribute how much to a city's pollution.

Complete Detailed Explanation

Part A — Understanding the Air Quality Index (AQI), In Depth

The AQI is a standardised scale that translates complex pollutant measurements into a single number and colour-coded category, making it easy for the public to understand health risk at a glance. India's AQI is calculated from eight pollutants, with the worst sub-index determining the overall reading.

AQI Range	Category	Health Implication
0–50	Good	Minimal impact
51–100	Satisfactory	Minor discomfort to sensitive people
101–200	Moderate	Breathing discomfort for sensitive groups (children, elderly, asthmatics)
201–300	Poor	Breathing discomfort for most people on prolonged exposure
301–400	Very Poor	Respiratory illness on prolonged exposure
401–500	Severe	Affects healthy people; serious impact on those with existing illness

Part B — The Ten Building Blocks of Module 4

This is the heart of the module. Each topic below is explained with its core meaning and its practical relevance for Indian citizens.

1. What is Air Pollution?

Air pollution is the presence of harmful substances in the atmosphere at concentrations that damage human health, ecosystems, and the climate — arising from both natural sources (dust storms, wildfires) and human activity (vehicles, industry, burning).

2. Types of Air Pollutants

Pollutants range from particulate matter (PM2.5, PM10) to gases like carbon monoxide, sulfur dioxide, and nitrogen oxides, along with volatile organic compounds and ground-level ozone — each with different health effects and sources.

3. Causes of Air Pollution

Major causes include vehicle emissions, industrial exhaust, burning of fossil fuels, crop residue/stubble burning, construction dust, and indoor sources like cooking with solid fuels or wood-burning stoves.

4. Effects on Human Health

Prolonged exposure to polluted air contributes to respiratory diseases (asthma, bronchitis), cardiovascular problems, reduced lung function, and increased risk of premature death. Children, the elderly, and those with pre-existing conditions are most vulnerable.

5. Indoor vs Outdoor Air Pollution

Outdoor pollution stems mainly from vehicles, industries, and construction, while indoor pollution arises from cooking smoke, poor ventilation, mould, and household chemicals — indoor air can be significantly more polluted than outdoor air in poorly ventilated spaces.

6. Air Quality Index (AQI)

The AQI is a standardised scale used to communicate how polluted the air currently is and the associated health risk, calculated from real-time monitoring of key pollutants across a growing national network of stations.

7. Pollution Prevention Methods

Prevention strategies include promoting clean fuels, enforcing vehicle emission standards, controlling industrial discharge, banning open burning of waste and crop residue, and increasing urban green cover to naturally filter air.

8. Green Transportation

Green transportation options — cycling, walking, public transport, carpooling, and electric vehicles — reduce vehicular emissions, ease traffic congestion, and lower a community's overall air pollution contribution.

9. Government Pollution Control Measures

India's National Clean Air Programme (NCAP) targets a 40% reduction in PM10 levels (or meeting the national standard) across 130 non-attainment cities, supported by Bharat Stage emission norms and pollution control board enforcement.

10. Personal Air Safety Practices

Individuals can protect themselves by checking the daily AQI before outdoor activity, using N95 masks on high-pollution days, avoiding outdoor exercise during smog, using air purifiers or indoor plants, and never burning waste at home.

Important Terminologies

Term	Simple Meaning
AQI	Air Quality Index — a standardised scale showing current pollution level and health risk
PM2.5	Fine particulate matter that can penetrate deep into the lungs and bloodstream
PM10	Coarser particulate matter, still small enough to be inhaled
NAAQS	National Ambient Air Quality Standards set by the CPCB
CAAQMS	Continuous Ambient Air Quality Monitoring Station
Non-Attainment City	A city that has failed to meet NAAQS over five consecutive years
Source Apportionment	A study identifying which sources contribute how much to a city's pollution
Smog	A visible haze formed by the combination of smoke and fog or ground-level ozone
Stubble Burning	The practice of burning leftover crop residue after harvest
Bharat Stage (BS) Norms	India's vehicle emission standards, similar to Euro norms

Step-by-Step Understanding

How to Respond to a High AQI Day

1. **Check** — Look up the current AQI for your area on an official app or website
2. **Assess** — Identify which category it falls into (Moderate, Poor, Very Poor, Severe)
3. **Protect** — Wear an N95 mask outdoors if AQI is Poor or worse, especially for sensitive groups
4. **Limit** — Reduce or avoid outdoor exercise and prolonged outdoor exposure on high-AQI days
5. **Ventilate Wisely** — Keep windows closed during peak pollution hours; use an air purifier indoors if available

How to Reduce Your Personal Contribution to Air Pollution

1. Maintain your vehicle regularly to keep emissions in check
2. Choose walking, cycling, public transport, or carpooling for short trips
3. Never burn leaves, garbage, or crop residue in the open
4. Support and use cleaner cooking fuel over solid fuels where possible
5. Report visible industrial or vehicular pollution violations to the authorities
6. Support urban green cover initiatives such as tree plantation drives

Real-Life Indian Examples

(Widely known public examples, described for educational awareness.)

- **Delhi's Winter Smog Crisis:** Every winter, Delhi and the wider National Capital Region experience a sharp spike in PM2.5 levels driven by a combination of vehicle emissions, stubble burning in neighbouring states, and unfavourable weather conditions that trap pollutants close to the ground.
- **Graded Response Action Plan (GRAP):** A staged emergency response system for Delhi-NCR that automatically triggers specific restrictions — such as construction bans or vehicle curbs — as AQI crosses defined thresholds.
- **PRANA Portal:** The CPCB's Portal for Regulation of Air-pollution in Non-Attainment cities tracks city-wise clean air action plans and fund utilisation for public transparency.
- **National Monitoring Expansion:** India's air quality monitoring network expanded to over 1,600 stations across 584 cities in 2025, a significant increase aimed at giving more cities real-time visibility into their own air quality (CREA, 2026).

Practical Scenarios

Scenario 1: The AQI in your city has crossed 300 ("Very Poor") for several consecutive days.

Limit outdoor activity, especially for children and the elderly, wear an N95 mask if you must go outside, and keep windows closed during peak pollution hours.

Scenario 2: A construction site near your home is generating visible dust without any covering or water spraying.

This is a common violation of dust-control norms. Report it to your local municipal authority or pollution control board, which can direct the site to use mandatory dust suppression measures.

Scenario 3: A neighbour is burning dry leaves and household waste in their backyard.

Open burning releases toxic pollutants and is prohibited under most municipal solid waste rules. Raise it directly if safe to do so, and report it if it continues.

Government Rules / Guidelines

- **Ministry of Environment, Forest and Climate Change (MoEFCC):** Launched and oversees the National Clean Air Programme (NCAP), India's national strategy to reduce particulate pollution across non-attainment cities.
- **Central Pollution Control Board (CPCB):** Sets and enforces National Ambient Air Quality Standards (NAAQS), operates the national air quality monitoring network, and runs the PRANA transparency portal.
- **State Pollution Control Boards (SPCBs):** Implement and enforce air quality regulations at the state and local level, including industrial emission compliance.
- **Commission for Air Quality Management (CAQM):** A statutory body coordinating air quality management across Delhi and the National Capital Region, including the Graded Response Action Plan (GRAP).

Laws & Policies

Law / Policy	What It Covers
Air (Prevention & Control of Pollution) Act, 1981	Regulates air pollutant emissions and establishes air quality standards and enforcement powers
Environment (Protection) Act, 1986	The umbrella law empowering government action against any activity, including air pollution, that harms the environment
National Clean Air Programme (NCAP), 2019	A national strategy targeting a 40% reduction in PM10 levels (or meeting NAAQS) across 130 non-attainment cities
Commission for Air Quality Management Act, 2021	Establishes the CAQM for coordinated air quality governance across Delhi-NCR
Bharat Stage (BS) Emission Norms	Regulate permissible vehicle exhaust emissions, aligned with international Euro standards
Graded Response Action Plan (GRAP)	A staged set of emergency measures triggered automatically as AQI crosses defined thresholds in Delhi-NCR

> *The National Clean Air Programme initially targeted a 20–30% reduction in PM10 by 2024 compared to 2017 levels; this was revised in 2022 to a more ambitious 40% reduction target (or meeting the national standard) by 2025–26.*

Important Statistics

Metric	Figure
Cities exceeding annual PM10 standard (2025)	190 out of 229 monitored cities
Most polluted Indian cities by annual PM2.5 (2025)	Byrnihat (100 µg/m³), Delhi (96 µg/m³), Ghaziabad (93 µg/m³)
National air quality monitoring stations (2025)	Over 1,600 (565 CAAQMS + 1,035 manual stations)
Cities covered by the monitoring network (2025)	584 cities across 28 states and 7 Union Territories
NCAP non-attainment cities under the programme	130 cities across 24 states/UTs
NCAP PM10 reduction target	40% reduction by 2025–26 (or meeting the 60 µg/m³ NAAQS)
NCR cities exceeding NAAQS (2025, of those monitored)	12 out of 14 cities with sufficient data

(Figures sourced from the Centre for Research on Energy and Clean Air (CREA) 'Tracing the Hazy Air' 2026 progress report on NCAP, and CPCB data; official portals should be checked for the latest updates.)

Best Practices

- Check the daily AQI before planning outdoor activities, especially for children and the elderly
- Maintain vehicles regularly to minimise emissions
- Choose public transport, carpooling, cycling, or walking where feasible
- Never burn leaves, garbage, or crop residue in the open
- Use N95 masks and air purifiers on high-pollution days
- Support and participate in local tree plantation and green cover initiatives

Common Mistakes

- Ignoring the AQI and continuing intense outdoor activity on high-pollution days
- Burning dry leaves or waste in the open, especially during winter
- Assuming indoor air is automatically safer without checking ventilation or indoor pollution sources
- Overlooking regular vehicle maintenance, which increases emissions over time
- Not reporting visible construction dust or industrial emission violations

Myths vs Facts

Myth	Fact
"Air pollution is only a problem in winter"	Levels fluctuate seasonally, but many Indian cities exceed safe standards for a significant part of the year, not just winter
"Staying indoors always means safe air"	Indoor air can be more polluted than outdoor air without proper ventilation, especially where solid fuels or poor air circulation are present
"Only industries and vehicles cause air pollution"	Construction dust, open burning, and household cooking fuel also contribute significantly to local air quality
"AQI is only relevant for people with existing health conditions"	Prolonged exposure at poor AQI levels affects healthy individuals too, particularly with sustained outdoor activity

Do's and Don'ts

Do	Don't
Check the AQI before outdoor activity	Don't ignore high AQI warnings, especially for children and the elderly
Maintain your vehicle regularly	Don't skip regular emission checks
Use public transport or carpool where possible	Don't default to private vehicles for every short trip
Report visible pollution violations	Don't burn leaves, garbage, or crop residue in the open
Support local tree plantation efforts	Don't assume individual actions don't add up collectively

Practical Tips

For Banking & Daily Commute

Prefer public transport, carpooling, or cycling for short trips • Keep vehicle engines properly maintained • Avoid unnecessary idling in traffic

For Families & Children

Check AQI before outdoor play or exercise • Keep N95 masks available for high-pollution days • Encourage indoor activity when AQI is Poor or worse

For Senior Citizens & Sensitive Groups

Limit outdoor exposure during high-pollution periods • Keep prescribed respiratory medication accessible • Use an air purifier indoors where feasible

Practical Checklist

■	I check the AQI regularly, especially before outdoor activity
■	I know what AQI category is considered unsafe for sensitive groups
■	I keep my vehicle well-maintained to reduce emissions
■	I never burn leaves, garbage, or crop residue in the open
■	I know which authority to contact to report a pollution violation
■	I have N95 masks available at home for high-pollution days
■	I have discussed air quality awareness with family or peers

Quick Revision Notes

- Air pollution = harmful gases and particulate matter contaminating the atmosphere
- AQI translates pollutant levels into an easy-to-read scale from Good to Severe
- PM2.5 and PM10 are the most closely tracked particulate pollutants in India
- NCAP targets a 40% reduction in PM10 (or meeting NAAQS) across 130 non-attainment cities
- CPCB, SPCBs, and CAQM are India's key air-quality regulatory bodies
- Report violations (open burning, uncontrolled dust, industrial smoke) to your local pollution control board

Key Takeaways

1. Air pollution affects nearly everyone in a region simultaneously — it cannot be entirely avoided by individual choices alone

2. Checking and responding to the daily AQI is a simple, effective personal safety habit

3. A significant share of air pollution stems from everyday, avoidable sources — open burning, poorly maintained vehicles, uncontrolled dust

4. India's monitoring infrastructure has expanded significantly, but sustained enforcement and citizen reporting remain essential

5. Small collective changes in transportation, waste burning, and reporting genuinely shape local air quality over time

Glossary

- **AQI** — Air Quality Index, a standardised scale for reporting pollution levels and health risk
- **CAAQMS** — Continuous Ambient Air Quality Monitoring Station
- **GRAP** — Graded Response Action Plan, staged emergency pollution-control measures for Delhi-NCR
- **NAAQS** — National Ambient Air Quality Standards set by the CPCB
- **Non-Attainment City** — A city that has failed to meet NAAQS over five consecutive years
- **PM2.5** — Fine particulate matter (2.5 micrometres or smaller) that penetrates deep into the lungs
- **PM10** — Coarser particulate matter (up to 10 micrometres) that can still be inhaled
- **Smog** — Visible haze formed from smoke, fog, or ground-level ozone
- **Source Apportionment** — A study identifying the relative contribution of different pollution sources
- **Stubble Burning** — Burning leftover crop residue after harvest, a major seasonal pollution source

Frequently Asked Questions

1. What is the Air Quality Index (AQI)?
A standardised scale from 0 to 500+ that translates pollutant concentrations into an easy-to-read category — Good, Satisfactory, Moderate, Poor, Very Poor, or Severe — along with health implications.

2. What is the difference between PM2.5 and PM10?
PM2.5 refers to fine particles 2.5 micrometres or smaller, which can penetrate deep into the lungs and bloodstream; PM10 refers to coarser particles up to 10 micrometres, which are still inhalable but less deeply penetrating.

3. Why is air quality worse in winter in northern India?
A combination of stubble burning in neighbouring states, vehicle and industrial emissions, and cooler, stagnant weather conditions traps pollutants closer to the ground during winter months.

4. Is indoor air always safer than outdoor air?
Not necessarily. Poor ventilation, cooking with solid fuels, and household chemicals can make indoor air more polluted than outdoor air in some cases.

5. What is the National Clean Air Programme (NCAP)?
A national strategy launched in 2019 targeting a 40% reduction in PM10 levels (or meeting NAAQS) across 130 non-attainment cities by 2025–26.

6. What is a 'non-attainment city'?
A city that has consistently failed to meet National Ambient Air Quality Standards over five consecutive years, requiring a dedicated clean air action plan.

7. What is GRAP?
The Graded Response Action Plan, a staged set of emergency measures — such as construction bans or vehicle restrictions — automatically triggered as AQI crosses defined thresholds in Delhi-NCR.

8. Which law regulates air pollution in India?
The Air (Prevention & Control of Pollution) Act, 1981, which established air quality standards and enforcement powers for pollution control boards.

9. Should I exercise outdoors when AQI is high?
It is best to limit or avoid prolonged outdoor exercise when AQI is in the Poor category or worse, especially for children, the elderly, and those with respiratory conditions.

10. How can I report an air pollution violation?
Contact your local municipal authority or State Pollution Control Board; the CPCB's PRANA portal also provides city-wise pollution action plan information.


JANGYAN

Official References

- Ministry of Environment, Forest and Climate Change (MoEFCC) — National Clean Air Programme (NCAP) information
- Central Pollution Control Board (CPCB) — National Ambient Air Quality Standards and monitoring data
- Commission for Air Quality Management (CAQM) — Graded Response Action Plan (GRAP) for Delhi-NCR
- Centre for Research on Energy and Clean Air (CREA) — 'Tracing the Hazy Air' NCAP progress reports
- Air (Prevention & Control of Pollution) Act, 1981 — India Code portal

Official Government Website Links

- Central Pollution Control Board: **https://cpcb.nic.in**
- PRANA Portal (NCAP tracking): **https://prana.cpcb.gov.in**
- Ministry of Environment, Forest and Climate Change: **https://moefcc.gov.in**
- Commission for Air Quality Management: **https://caqm.nic.in**
- Press Information Bureau: **https://pib.gov.in**

Conclusion

Clean air is not a luxury — it is the most basic requirement for public health, and one that no individual can fully secure alone. Yet every citizen's daily choices — how they commute, whether they burn waste, whether they report a violation — feed directly into the air an entire community breathes. Understanding the AQI and acting on it is one of the simplest, most direct ways to protect yourself and those around you.

Continue your Environment & Civic Responsibility journey with JanGyan. Stay alert, stay informed, and remember: clean air is a shared responsibility, breath by breath.

JanGyan — Learn Without Limits
Module 4 of the Environment & Civic Responsibility Category